

RUCHIT PAHADIA

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SUMMARY

Final-year CSE student and ML engineer with two industry internships and a Samsung-led 6-month Advanced AI program. Ship end-to-end ML, deep learning, computer vision, and NLP systems — from data pipelines to edge deployment. Track record of measurable impact: 91% detection accuracy, 47% lower prediction error. Targeting ML Engineer, Data Science, and AI Developer roles.

TECHNICAL SKILLS

Languages: Python, Java, C, SQL

ML / Deep Learning: TensorFlow, TensorFlow Lite, PyTorch, Keras, Scikit-learn, NumPy, Pandas, Matplotlib, Seaborn, CI/CD Pipelines

Computer Vision: OpenCV, Deep Learning Object Detection, SORT Tracker, ONNX, OCR

NLP: Hugging Face Transformers, BERT, spaCy, NLTK

Edge AI & Deployment: Model Optimisation, On-Device Inference, TensorFlow Lite, ONNX, Raspberry Pi, Arduino

Databases & Tools: MySQL, Git, GitHub, VS Code, REST APIs, Unity, Blender

Core Concepts: Supervised Learning, Feature Engineering, Model Deployment, Anomaly Detection, IoT Systems

EXPERIENCE

IoT & Advanced AI Intern

Mar 2026 – Present

Ganaka Praudyogikee Tech Solutions

Bengaluru, Karnataka

- Building AI-integrated IoT pipelines for real-world automation, reducing manual intervention in operational processes.
- Designing and testing deep learning models for sensor-data interpretation and anomaly detection on edge hardware.
- Deploying optimised models to embedded systems (Raspberry Pi) with custom preprocessing pipelines and on-device inference tuning.

Data Science Intern

Jun 2025 – Jul 2025

BNMIT in association with E2E Technologies

Bengaluru, Karnataka (On-site)

- Built and fine-tuned supervised ML models (Random Forest, XGBoost), achieving ~85% accuracy on classification tasks.
- Engineered end-to-end preprocessing pipelines (imputation, normalisation, feature selection) on datasets of 50,000+ records.
- Evaluated models using cross-validation, precision-recall, and AUC-ROC metrics; presented findings to senior stakeholders.

Advanced AI Program — Samsung Innovation Campus

Aug 2025 – Feb 2026

Exclusive 6-Month Offline Program by Samsung officials — highly selective intake

120+ hours (On-Site)

- Selected through competitive screening for Samsung's flagship 6-month Advanced AI cohort — 120+ hours of in-person ML, Deep Learning, neural networks, and production-grade model architectures, taught by Samsung R&D officials.

PROJECTS

Patient Handoff System (Ongoing) | Python, Hugging Face Transformers, spaCy, Flask

2025 – Present

- Architecting an end-to-end clinical-handoff platform with a transformer-based NLP summary generator and interactive chatbot, plus full data pipelines (ingestion, entity extraction, summarisation, dialogue) producing structured handoff reports from raw clinical notes.

AI-Based Traffic Violation Detection System | Python, PyTorch, OpenCV, SORT Tracker, Custom OCR

2025

- Built an end-to-end computer vision and deep learning pipeline detecting multiple violations — helmet absence, triple riding, and wrong-lane driving — from live traffic video footage.
- Integrated SORT-based vehicle tracking and a custom OCR model to link violations to licence plates, logging plate/type/timestamp — achieving ~91% vehicle-detection, ~87% plate-recognition accuracy, and cutting false positives by 30%.

DealerXP | React, FastAPI, PostgreSQL, Redis, Docker

2026

- Built a full-stack gamification layer for automotive dealership CRM/DMS pipelines, mapping lead-to-delivery event data into XP, streaks, badges, and department leaderboards via a FastAPI backend and React/Vite frontend.
- Engineered a scoring and anti-gaming engine that filters 2,000+ raw operational events into 20 milestone-based actions, applying rate-capping, collusion gating, and behavioral anomaly detection to block point manipulation.

Zenith — Voice Assistant | Python, SpeechRecognition, pyttsx3

2024

- Created a voice assistant supporting app launching, web search, calendar queries, and speech-to-text/TTS with ~95% command-recognition accuracy.

EDUCATION

B.E. in Computer Science Engineering

BNM Institute of Technology • 2023 – Present

CGPA: 8.03 / 10

Senior Secondary, CBSE (PCM + CS)

Presidency School Bangalore South • 2021 – 2023

Aggregate: 83%

ACHIEVEMENTS & CERTIFICATIONS

2nd Place, BNMIT Ideathon (2024) — presented the Zenith Voice Assistant project against college-wide competition.

Certifications: NPTEL — AI & ML (2024) • MongoDB University (2024) • IoT Using Python, IIT Bombay (2024)